

Editors-in-Chief
Journal of Machine Learning Research

Re: Grokking the Discrete Logarithm: Scaling Across Model Capacity and Mechanistic Analysis

Dear Editors-in-Chief,

Please consider the enclosed manuscript, “Grokking the Discrete Logarithm: Scaling Across Model Capacity and Mechanistic Analysis,” for publication in the *Journal of Machine Learning Research*.

The paper studies the discrete logarithm problem as a controlled machine-learning benchmark for delayed algorithmic generalization. To the best of my knowledge, it provides the first systematic demonstration of grokking on the full variable-base DLP mapping, together with a capacity sweep from 90k to 14.4M parameters, mechanistic spectral analysis, symbolic regression of hidden representations, prime-order controls, and generalization to generators excluded from training.

Previous or overlapping publication. The work has not been published in another journal, conference, or workshop. A non-peer-reviewed preprint of the submitted manuscript is publicly available on Zenodo at <https://doi.org/10.5281/zenodo.22813397> (DOI: 10.5281/zenodo.22813397). The accompanying code, data, split manifests, analysis pipelines, and reproducibility materials are publicly available at <https://github.com/VesterlundCoder/dlp-grokking>.

Authorship. I am the sole author of this manuscript, so no co-author consent is applicable. I confirm that the submitted author list is final for the review period.

Conflicts of interest. I confirm that no conflicts of interest apply with any of the proposed Action Editors or reviewers listed below.

Suggested Action Editors, subject to conflict-of-interest verification.

- Joan Bruna – deep learning theory, signal processing, statistics.
- Samy Bengio – deep learning, representation learning.
- Mehryar Mohri – learning theory, spectral learning.
- Kilian Weinberger – deep learning, representation learning.

Suggested reviewers, subject to conflict-of-interest verification.

- Neel Nanda – grokking and mechanistic interpretability.
- Ziming Liu – grokking and representation learning.
- Vikrant Varma – circuit efficiency and grokking.
- Miles Cranmer – symbolic regression and interpretable machine learning.
- Huu Danh Nguyen – discrete-log/character representations in transformer arithmetic.

Keywords. grokking; algorithmic generalization; discrete logarithm; mechanistic interpretability; representation learning.

Use of AI-assisted tools. For transparency, AI-assisted tools were used for programming assistance, code generation, language drafting, and editorial support. The research questions, hypotheses, experimental design, experiment selection, mathematical reasoning, code review, test execution and validation, result interpretation, proofreading, and final editing were carried out or directly verified by the author. All scientific claims and conclusions are the author’s responsibility.

Funding and third-party support. Compute resources were provided by the LUMI-G supercomputer (EuroHPC JU) under project 465003364. Additional experiments were run on local Apple MPS hardware. No other third-party funding or support was received.

Thank you for your consideration.

Sincerely,

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